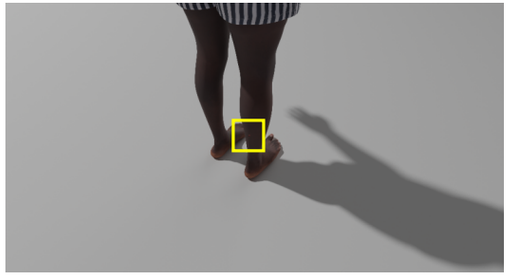
Full pipeline on a rendered patient leg (Syn3DWound, CC BY-NC-SA 4.0): photographs → wound → surface → target → honeycomb → staged fill → scaffold on the body

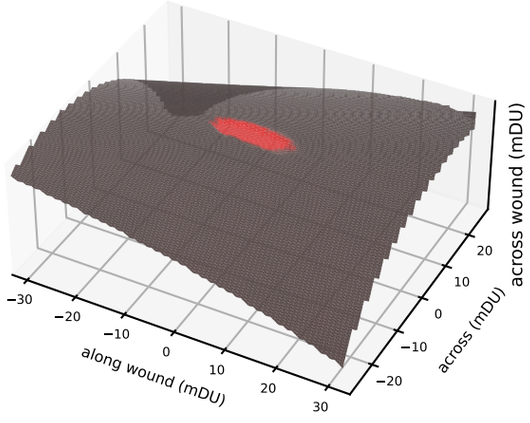
1 OBSERVE: one of 40 photographs
(rendered leg, image_000.png)



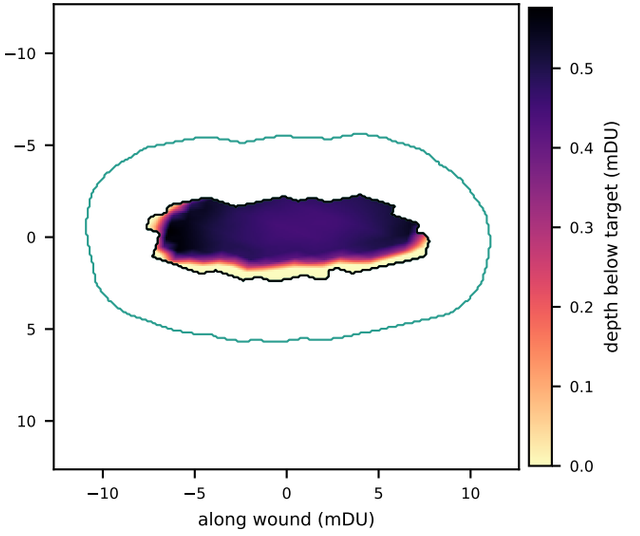
2 IDENTIFY (2D): the dataset's wound mask
(yellow outline; zoom into the yellow box)



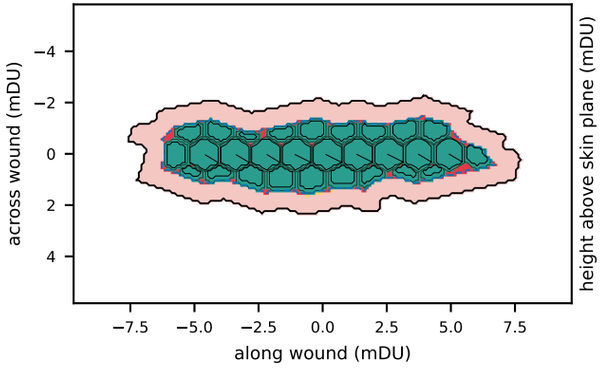
2 IDENTIFY (3D): masks from 40 views voted
onto the leg mesh (153 wound vertices, red)



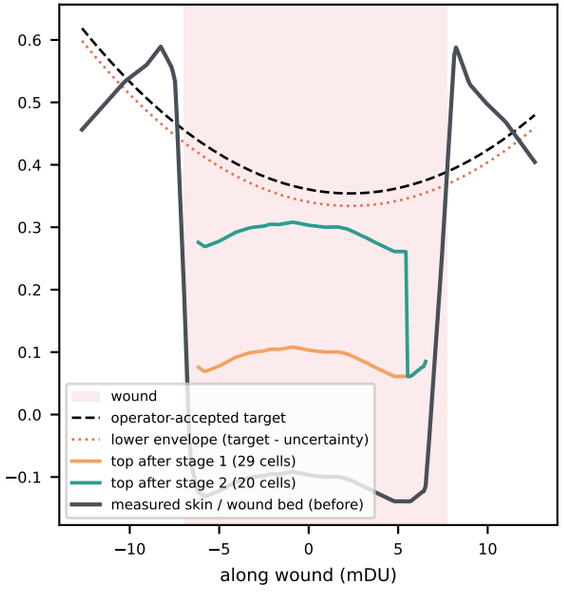
3-4 SURFACE + TARGET: how far the wound bed lies
below the operator-accepted target (green = healthy ring)



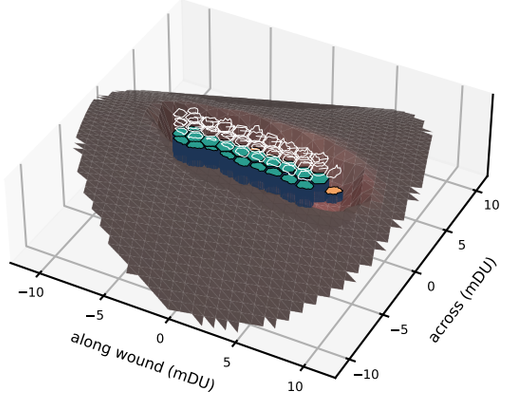
5 HONEYCOMB COVERAGE: 29 cells cover
94.2% of safe area, 50.8% of wound;
red = safe area left uncovered, pink = edge margin



6 STAGED FILL: slice through the printed cells
stop rule fired: NO_SAFE_SUPPORTED_REGION



7 PLACE (3D): honeycomb scaffold sitting in the
wound on the leg (heights x4 for visibility)



OUTPUT: the planned scaffold projected back onto the patient photograph (plan only; printing disabled)

